PRINCETON UNIVERSITY

Two Scalable Distributed Strategies

for Nonlinear Programming in Power Systems

AI4OPT Seminar at Georgia Institute of Technology
Xinliang Dai | xinliang.dai@princeton.edu



Power Systems Overview

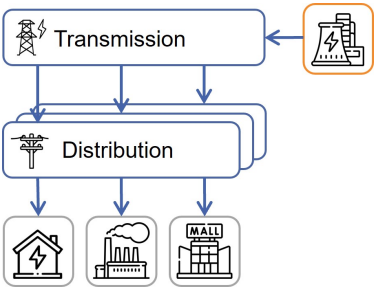


Figure 1: Traditional Power Systems

- **Traditional Power Systems**
  - **Centralized Generation**
  - **Unidirectional Power Flow**
  - **Predictable Demand**

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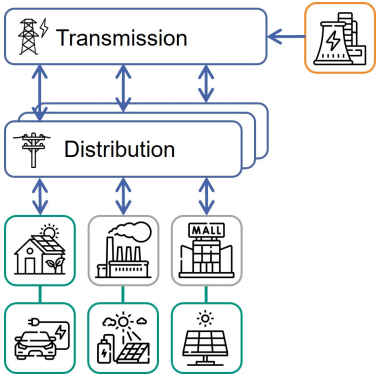


Figure 1: Modern Power Systems

- **Traditional Power Systems**
  - **Centralized Generation**
  - **Unidirectional Power Flow**
  - **Predictable Demand**
- **Power Systems during Energy Transition**
  - **Distributed Energy Resources (DERs)**
  - **Bidirectional Power Flow**
  - **Real-Time Computation**
  - **Cybersecurity & Privacy Concerns**

Nonlinear Programs in Power Systems

Alternating Current Optimal Power Flow Problems

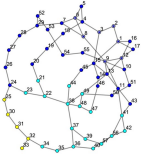


Figure 2: IEEE 57-bus system

- Optimization Problem:

$$\min_{x} \quad f(x) \tag{1a}$$

$$\text{s.t.} \quad c(x) \leq 0 \tag{1b}$$

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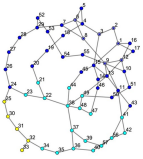


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  - minimize totoal generation/operation cost
  - subject to system models including:
    - nodal power balance
    - voltage magnitude limits
    - transmission line power limits,
    - generator power limit

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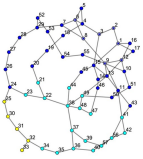


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- Applications in power systems
  - Optimal Economic Dispatch
  - N-1 security-constrained OPF
  - Shapley value-based congestion management
  - Power network expension

Graph-Based Distributed Optimization

An Example in Power System Network

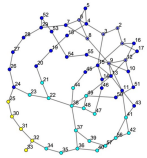


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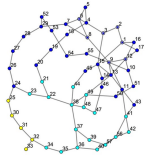


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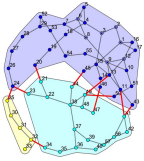


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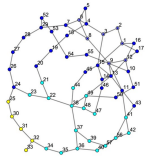


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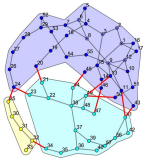


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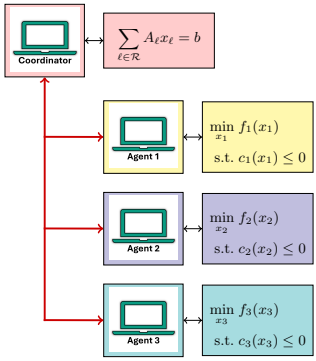


Figure 2: Distributed systems for distributed optimization

Different Approaches in Power Systems Applications

Single-Threaded vs. Multi-Threaded vs. Distributed Optimization

Table 1: Centralized vs. Distributed Coordination

	Centralized (Single-Threaded)	Centralized (Multi-Threaded)	Distributed
Formulation	Original		Complex Reformulation
Parallelization	No	Linear solver	Graph
Processing	Full data centralization		Local data processing
Adaptability	Slow response		Fast local reconfiguration
Resiliency	Single-point vulnerability		Failure localized
Scalability	- -		+ +
Algorithms	Well-established		Potentially more complex

Content

1. Introduction

2. Incremental Advances in Distributed Optimization for Nonlinear Programming

3. Barrier Augmented Lagrangian-based Alternating Direction Inexact Newton (BALADIN)

4. Flexibility Aggregation for Coordinating Integrated Transmission-Distribution (ITD) Systems

5. Conclusion & Future Work

Literature Review: Distributed Optimization for AC OPF

Table 2: Distributed optimization for AC OPF

Ref.	Alg.	Convergence	Speed	Accuracy	Execution	Scalability
[2], [3], [4]	ADMM	-	+	+	Sequential	10^2
[5]	ADMM	-	+	+	Sequential	10^2
[6]	ADMM	-	+	+	Sequential	10^3
[7]	ADMM	-	+	+	Sequential	10^4
[8]	ADMM	Global	+	+	Parallel	10^4
[9], [10]	ALADIN	Global	+++	+++	Sequential	10^2
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ADMM: Alternating Direction Method of Multipliers

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ADMM vs. ALADIN

Methodological Differences

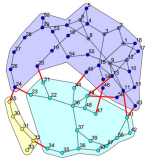


Figure 4: System with 3 regions [1]

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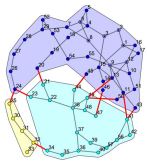


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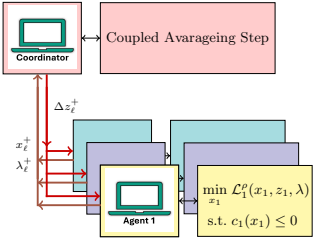


Figure 5: ADMM Framework

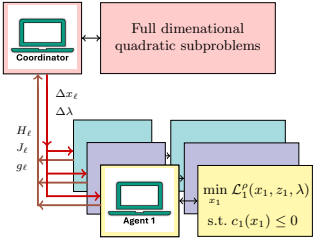


Figure 6: ALADIN Framework

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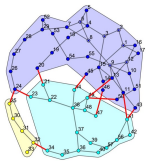


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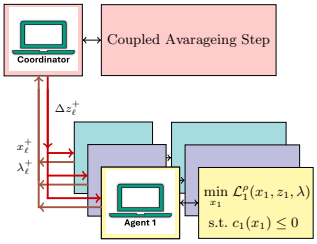


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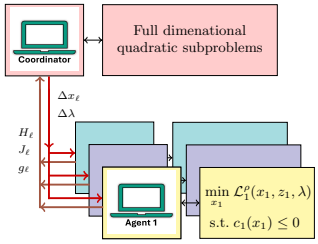


Figure 6: ALADIN Framework

Table 3: Algorithm Comparison: ADMM vs. ALADIN

	ADMM	ALADIN
Local Agents	Augmented Lagrangian Method	Augmented Lagrangian Method
Coordinator	Gradient Methods	Newton Method
Communication	Vectors	Matrices

[12] B. Houska et al., "An augmented lagrangian based algorithm for distributed nonconvex optimization," *SIAM Journal on Optimization*, 2016

ADMM vs. ALADIN

Numerical Performance: Nonconvex Case

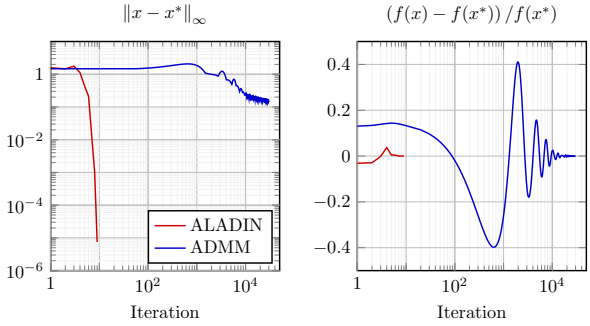


Figure 7: Iterative Comparison: ADMM vs. ALADIN [13]

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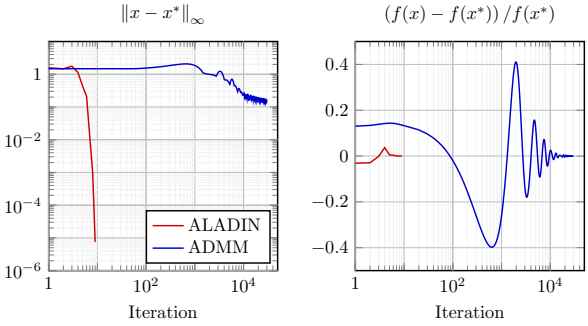


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Table 4: ADMM vs. ALADIN for Nonconvex Problems

	ADMM	ALADIN
Convergence Guarantees	NaN	Global
Convergence Rate	+	+++
Solution Accuracy	+	+++
Communication Overhead	-	- - -
Per Iteration Computation	-	- - -
Scalability	limited	?

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Challenges of ALADIN

Limited Scalability by Inequality Constraints

- Problems without inequality constraints

Table 5: ALADIN for AC power flow problems [14], [15]

Buses	Regions	Coupling	ALADIN [s]	Centralized [s]
53	3	5	0.011	0.004
418	2	8	0.028	0.014
2708	2	30	0.109	0.051
4662	5	130	0.228	0.129
10224	13	242	0.591	0.257

AC power flow problems have no inequality constraints

[14] **X. Dai** et al., "Rapid scalable distributed power flow with open-source implementation," in *9th IFAC Conference on Networked Systems*, 2022

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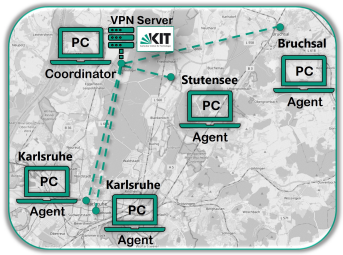
AC power flow problems have no inequality constraints

- Problems with inequality constraints
  - Inequality constraints **on the boundary** are included in **active-set** and treated as equality constraints.
  - Iteratively updates the **active-set** by **adding/removing constraints** based on local solutions.
  - A good initial estimate of the active set significantly improves scalability [16].

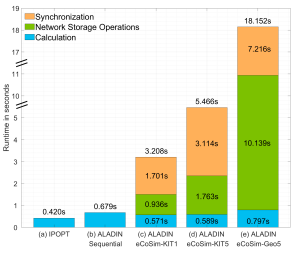
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Challenges of ALADIN

Network Topology Leakage via Jacobian Sparsity Patterns



(a) Distributed Computation



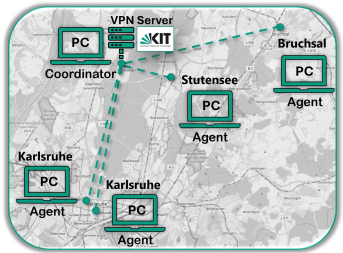
(b) TSOs Coordination

Figure 8: Geographically distributed computation in Karlsruhe area [17]

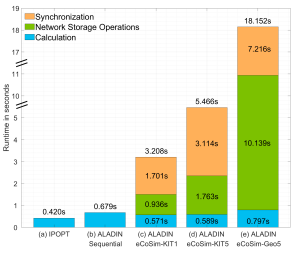
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Network Topology Leakage via Jacobian Sparsity Patterns



(a) Distributed Computation



(b) TSOs Coordination

Figure 8: Geographically distributed computation in Karlsruhe area [17]

- Subproblems should be solved within comparable time windows to reduce synchronization time.
- Reducing data exchange is crucial for cost-effective communication infrastructure.
- Sparsity patterns in full-dimensional QP models may reveal network topology

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Challenges of ALADIN

Network Topology Disclosure by Sparsity Pattern of Jacobian Matrix

- Power nodal balance at bus $\{1\}$:

$$c_1(x_1, x_2, x_3) = \underbrace{s_{21}(x_1, x_2) + s_{31}(x_1, x_3)}_{\text{power flow from neighboring buses}} - \underbrace{s_1(x_1)}_{\text{from devices}} = 0$$

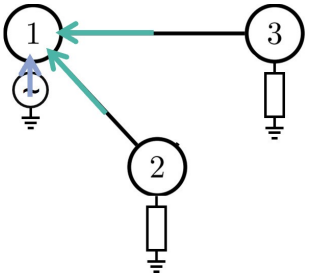


Figure 9: 3-bus example network

Challenges of ALADIN

Network Topology Disclosure by Sparsity Pattern of Jacobian Matrix

- Power nodal balance at bus $\{1, 2, 3\}$:

$$c_1(x_1, x_2, x_3) = 0$$
$$c_2(x_1, x_2) = 0$$
$$c_3(x_1, x_3) = 0$$

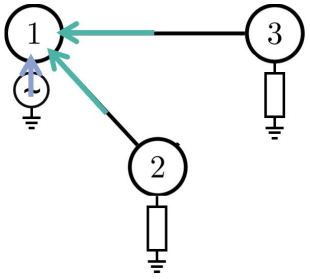


Figure 9: 3-bus example network

[18] **X. Dai** et al., "Distributed AC optimal power flow: A scalable solution for large-scale problems,", under review, 2026, [arXiv:2503.24086]

Challenges of ALADIN

Network Topology Disclosure by Sparsity Pattern of Jacobian Matrix

- Power nodal balance at bus $\{1, 2, 3\}$:

$$c_1(x_1, x_2, x_3) = 0$$
$$c_2(x_1, x_2) = 0$$
$$c_3(x_1, x_3) = 0$$

- Sparsity patten of the Jacobian matrix:

$$\begin{array}{l} \\ \text{constraints at bus } 1 \\ \text{constraints at bus } 2 \\ \text{constraints at bus } 3 \end{array} \begin{array}{c} \begin{array}{ccc} \text{bus } 1 & \text{bus } 2 & \text{bus } 3 \end{array} \\ \begin{pmatrix} \frac{\partial c_1}{\partial x_1} & \frac{\partial c_1}{\partial x_2} & \frac{\partial c_1}{\partial x_3} \\ \frac{\partial c_2}{\partial x_1} & \frac{\partial c_2}{\partial x_2} & 0 \\ \frac{\partial c_3}{\partial x_1} & 0 & \frac{\partial c_3}{\partial x_3} \end{pmatrix} \end{array}$$

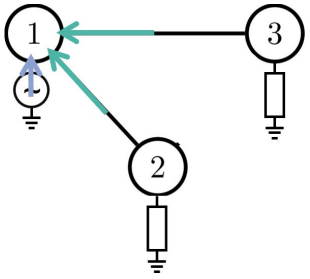


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Summary

Ref.	Alg.	Convergence	Speed	Accuracy	Execution	Commu.	Scalability	Problems
[2], [3], [4], [5], [6], [7]	ADMM	-	+	+	Sequential	-	10^3	OPF
[8]	ADMM	Global	+	+	Parallel	-	10^4	OPF
[9], [10], [11]	ALADIN	Global	+++	+++	Sequential	- - -	10^2	OPF
[13]	ALADIN	Global	+++	+++	Sequential	- - -	10^2	OPF
[14], [15]	ALADIN	Global	+++	+++	Sequential	- - -	10^4	PF
[16]	ALADIN	Global	+++	+++	Sequential	- - -	10^3	OPF-CR
[17]	ALADIN	Global	++	+++	Parallel	- - -	10^2	OPF

CR: Convex relaxation to radial network

- Challenges of standard ALADIN
  - scalability limited due to active-set method
  - communication overhead
  - Data privacy

[13] J. Zhai[§], **X. Dai**[§] et al., "Distributed optimal power flow for VSC-MTDC meshed AC/DC grids using ALADIN," *IEEE Trans. Power Syst.*, 2022, [§] **equally**
[14] **X. Dai** et al., "Rapid scalable distributed power flow with open-source implementation," in *9th IFAC Conference on Networked Systems*, 2022
[15] Y. Jiang[§], **X. Dai**[§] et al., "Global ALADIN for distributed nonconvex optimization,", 2026, under review, [§] **equally**
[16] **X. Dai** et al., "Advancing distributed AC optimal power flow for integrated transmission-distribution systems," *IEEE Trans. Netw. Sci. Eng.*, 2025
[17] **X. Dai** et al., "Ensuring data privacy in AC optimal power flow with a distributed co-simulation framework," *Electric Power Systems Research*, Sepecial issue from Power System Computation Conference (PSCC), 2024

Content

1. Introduction

2. Incremental Advances in Distributed Optimization for Nonlinear Programming

3. Barrier Augmented Lagrangian-based Alternating Direction Inexact Newton (BALADIN)

4. Flexibility Aggregation for Coordinating Integrated Transmission-Distribution (ITD) Systems

5. Conclusion & Future Work

Interior Point Method (IPM)

- Original NLP:

$$\min_{x} \quad f(x) \tag{5a}$$
$$\text{s.t.} \quad c^{\mathrm{E}}(x) = 0, \tag{5b}$$
$$c^{\mathrm{I}}(x) \leq 0, \tag{5c}$$

[19] J. Nocedal et al., *Numerical optimization*. Springer Science & Business Media, 2006

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- Barrier Subproblems:

$$\begin{aligned} \min_{x,s} \quad & f(x) - \mu \sum_{m \in \mathcal{C}^{\mathrm{I}}} \ln(s^{(m)}) && \text{(6a)} \\ \text{s.t.} \quad & c^{\mathrm{E}}(x) = 0 \qquad \mid \gamma, && \text{(6b)} \\ & c^{\mathrm{I}}(x) + s = 0 \quad \mid \kappa, && \text{(6c)} \end{aligned}$$

with barrier parameter μ and slack variable s

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- Solutions to barrier subproblems (central path)

$$x^\star(\mu) = \min_{x} \ (6a), \quad \text{s.t. } (6b) - (6c).$$

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Lemma (Convergence of the Central Path [19])

As the barrier parameter μ approaches zero with $s, \kappa \geq 0$, central path $x^\star(\mu)$, i.e., the solution to (6), converges to the solution of the original problem (5).

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Remark (Practical IPM Implementation)

Algorithmically, an IPM achieves convergence through an iterative two-phase process. For a fixed $\mu > 0$, an inner loop (e.g., using SQP-type steps with globalization) converges to the central path point $x^\star(\mu)$. An outer loop then drives $\mu \to 0$, generating a sequence of points converge to the optimal solution of the original problem.

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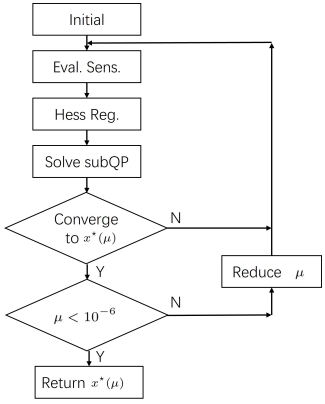


Figure 10: Flow Chart for Interior Point Method

Lemma (Convergence of the Central Path [19])

As the barrier parameter μ approaches zero with $s, \kappa \geq 0$, central path $x^\star(\mu)$, i.e., the solution to (6)*, converges to the solution of the original problem* (5)*.*

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Overview of Barrier ALADIN

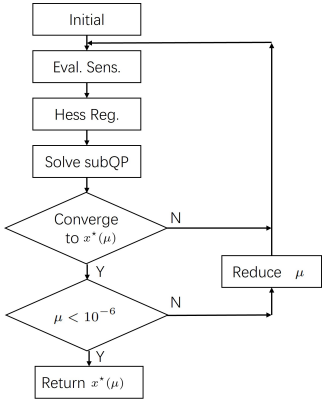


Figure 11: Flow Chart for Interior Point Method

- Barrier & ALADIN

Overview of Barrier ALADIN

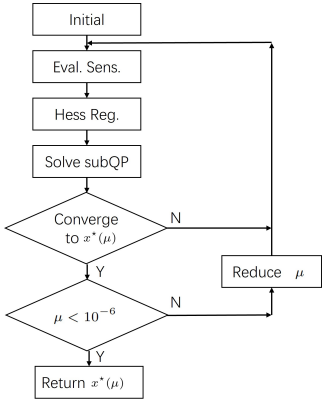


Figure 11: Flow Chart for Interior Point Method

- Barrier & ALADIN
  - Interior-point methods (IPM) to smooth inequality constraints

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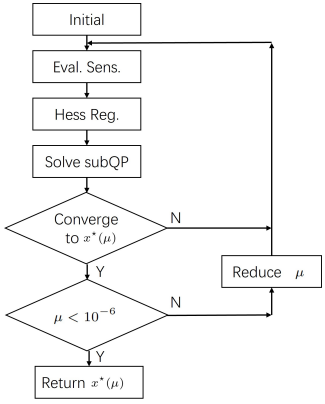


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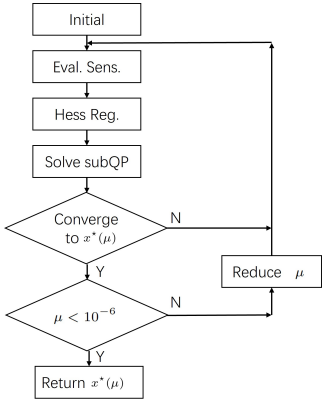


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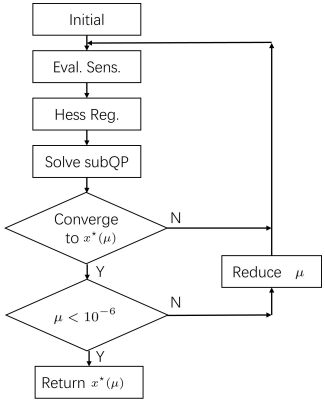


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  - Distributed Inertia Control (Hess. Reg)

Distributed Reformulation of Barrier Subproblems

- Original Problem:

$$\min_{x} \sum_{\ell \in \mathcal{R}} f_\ell(x_\ell) \tag{5a}$$

$$\text{s.t.} \sum_{\ell \in \mathcal{R}} A_\ell x_\ell = b \tag{5b}$$

$$c_\ell^{\mathsf{E}}(x_\ell) = 0,\ c_\ell^{\mathsf{I}}(x_\ell) \leq 0 \qquad \forall \ell \in \mathcal{R} \tag{5c}$$

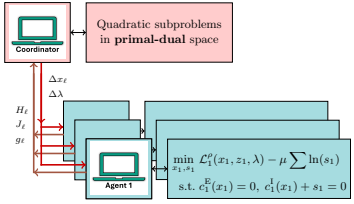


Figure 12: Primal-dual ALADIN framework for barrier problems (6)

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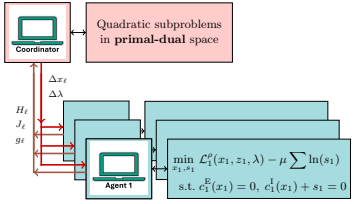


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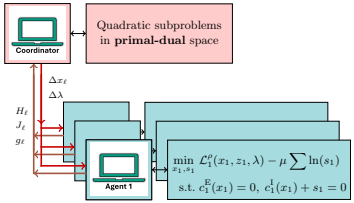


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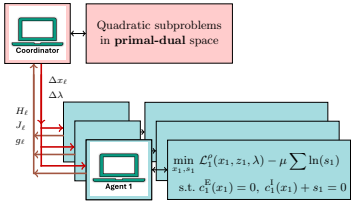


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Condensing Perturbed KKT

- Full-dimensional Newton step:

$$\begin{bmatrix} H_1 & J_1^\top & & & & & A_1^\top \\ J_1 & & & & & & \\ & & H_2 & J_2^\top & & & A_2^\top \\ & & J_2 & & & & \\ & & & & H_3 & J_3^\top & A_3^\top \\ & & & & J_3 & & \\ A_1 & & A_2 & & A_3 & & \end{bmatrix} \begin{bmatrix} \Delta x_1 \\ \Delta \kappa_1 \\ \Delta x_2 \\ \Delta \kappa_2 \\ \Delta x_3 \\ \Delta \kappa_3 \\ \Delta \lambda \end{bmatrix} = - \begin{bmatrix} g_1 \\ 0 \\ g_2 \\ 0 \\ g_3 \\ 0 \\ r \end{bmatrix}$$

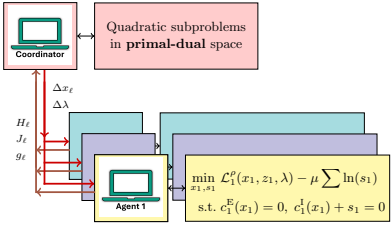


Figure 13: Dual ALADIN framework for barrier problems (6)

Condensing Perturbed KKT

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- Newton-Step with **Schur complement**

$$\sum_{\ell \in \mathcal{R}} W_\ell \, \Delta\lambda = -r + \sum_{\ell \in \mathcal{R}} h_\ell$$

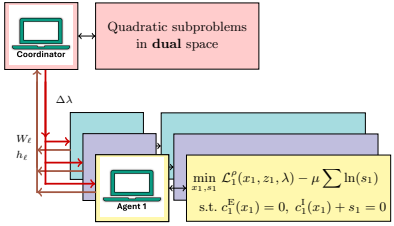


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Remark (Parallel Condensing)

$$W_\ell = -\begin{bmatrix} A_\ell & 0 \end{bmatrix} \begin{bmatrix} H_\ell & J_\ell^\top \\ J_\ell & \end{bmatrix}^{-1} \begin{bmatrix} A_\ell^\top \\ 0 \end{bmatrix}$$

$$h_\ell = -\begin{bmatrix} A_\ell & 0 \end{bmatrix} \begin{bmatrix} H_\ell & J_\ell^\top \\ J_\ell & \end{bmatrix}^{-1} \begin{bmatrix} g_\ell \\ 0 \end{bmatrix} + A_\ell x_\ell$$

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- Newton-Step with **Schur complement**

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- Communication: W_ℓ, h_ℓ and $\Delta\lambda$

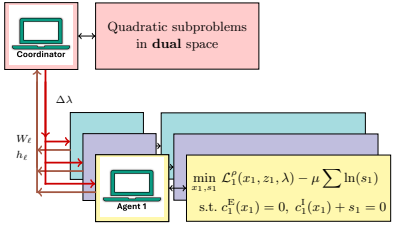


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Inertia Condition for Nonconvex QP Subproblems

Definition (Inertia [20])

The inertia of a symmetric matrix K is the triple

$$\text{inertia}(K) = (n^+, n^-, n^0),$$

where n^+, n^-, and n^0 denote the number of **positive**, **negative**, and **zero** eigenvalues of K, respectively.

[20] N. I. Gould, "On practical conditions for the existence and uniqueness of solutions to the general equality quadratic programming problem," *Mathematical Programming*, 1985

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Theorem (Inertia Condition for QP [20, Thm. 2.1])

Let K be a regular KKT matrix as

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$$\text{inertia}(K) = (n^x, \; n^c, \; 0) \tag{7}$$

$Z^\top H Z$ is positive definite, and the corresponding **QP** *subproblems has a* **strict minimizer**.

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Inertia Condition for Nonconvex QP Subproblems

Definition (Inertia [20])

The inertia of a symmetric matrix K is the triple

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where n^+, n^-, and n^0 denote the number of **positive**, **negative**, and **zero** eigenvalues of K, respectively.

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- Question: how to monitor the inertia in distributed approaches?

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Distributed Inertia Condition

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Theorem (Haynsworth Inertia Additivity Formula [21, Thm. 1])

Given that

$$K = \begin{bmatrix} K_{11} & K_{21}^\top \\ K_{21} & K_{22} \end{bmatrix}$$

is a Hermitian matrix and let K_{11} be the nonsingular and let $\Lambda = K_{22} - K_{21} K_{11}^{-1} K_{21}^\top$ be the Schur complement of K_{11} in K. Then

$$\text{inertia}(K) = \text{inertia}(\Lambda) + \text{inertia}(K_{11}).$$

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Distributed Inertia Condition

- Full-dimensional KKT:

$$K = \begin{bmatrix} \hat{H}_1 & & & & \hat{A}_1^\top \\ & \hat{H}_2 & & & \hat{A}_2^\top \\ & & \hat{H}_3 & & \hat{A}_3^\top \\ \hat{A}_1 & \hat{A}_2 & \hat{A}_3 & & \end{bmatrix}$$

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Corollary (Distributed Inertia Condition [18, Col. 1])

Let K be the regular KKT matrix, i.e., and let the Jacobian $\hat{J} = \begin{bmatrix} J^\top, A^\top \end{bmatrix}^\top$ be of full-row rank. Then, if dual Hessian W satisfies the inertia condition

$$\text{inertia}(K) = \text{inertia}(W) + \sum_{\ell \in \mathcal{R}} \text{inertia}(\hat{H}_\ell) = (\sum_\ell n_\ell^x,\ n^\lambda + \sum_\ell n_\ell^c,\ 0), \quad (9)$$

H is positive definite on the nullspace of the Jacobian $\bar{J}$ and the equivalent QP subproblem has a strict local minimizer.

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Criteria for Optimal Graph-based Decomposition

- Equitable Partitioning
  - Ensures subproblems are of similar size and computational effort.
  - Prevents idle time and enhances synchronization.

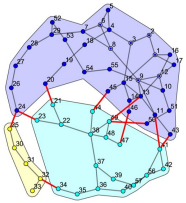


Figure 14: IEEE 57-bus system with 3 regions

Remark (Condensing)

$$W_\ell = -\begin{bmatrix} A_\ell & 0 \end{bmatrix} \begin{bmatrix} H_\ell & J_\ell^\top \\ J_\ell & \end{bmatrix}^{-1} \begin{bmatrix} A_\ell^\top \\ 0 \end{bmatrix}$$

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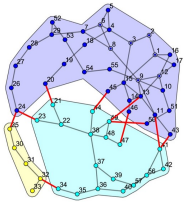


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- Trade-off by increasing agents
  - Reduces the size of decoupled subproblems.
  - Increases the computational burden in condensing.
  - Expands the dimension of QP subproblems in the dual space.

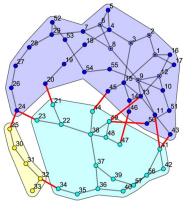


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Large-Scale Benchmark: Centralized vs. Distributed

Table 6: PGLib-OPF Benchmark*

Buses	Regions	Density	Mode	BALADIN [s]	IPOPT [s]
9,241	47	15.91%	STD	6.22	10.68
			API	5.09	19.11
			SAD	7.07	13.94
30,000	47	8.36%	STD	27.05	63.38
			API	30.80	73.96
			SAD	91.59	126.23
78,484	47	6.37%	STD	80.35	199.05
			API	83.01	1000+
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193,000	47	0.97%	STD	37.35	178.33

STD: standard; **API**: active power increased; **SAD**: small angle difference

- Setting
  - Parallel computing
  - Distributed memory
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Large-Scale Benchmark: Centralized vs. Distributed

Table 6: Practical Iterative Communication Effort

Buses	Density	ADMM	ALADIN	BALADIN
13,659	9.64%	0.427 MB	2.388 MB	0.682 MB
24,464	11.97%	0.635 MB	4.276 MB	2.104 MB
78,484	6.37%	2.016 MB	13.731 MB	5.791 MB
193,000	0.97%	4.999 MB	20.226 MB	0.888 MB

Note: practical communication efforts for large-scale AC OPF benchmarks are measured in MegaBytes, assuming the use of single-precision floating-point

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- Test cases
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Summary

Table 7: Distributed optimization algorithms for AC PF/OPF

Ref.	Alg.	Convergence	Speed	Accuracy	Execution	Commu.	Scalability	Problems
[2], [3], [4], [5], [6], [7]	ADMM	*NaN*	+	+	Sequential	-	10^3	OPF
[8]	ADMM	Global	+	+	Parallel	-	10^4	OPF
[9], [10], [11], [12], [13], [14], [16]	ALADIN	Global	+++	+++	Sequential	- - -	10^2	OPF
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- Methodology
- Take Away

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  - IPM outer loop
  - ALADIN for solving barrier problems
  - Distributed processing
    - Automatic differentiation (AD)
    - Derivatives condensing
    - Inertia control
    - Less communications
- Take Away

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  - Distributed processing
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    - Derivatives condensing
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  - Smooth nonconvex nonlinear programs

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[17]	ALADIN	Global	++	+++	Parallel	- - -	10^2	OPF
[18]	BALADIN	Global	++++	+++	Parallel	- -	10^5	OPF

- Methodology
  - IPM outer loop
  - ALADIN for solving barrier problems
  - Distributed processing
    - Automatic differentiation (AD)
    - Derivatives condensing
    - Inertia control
    - Less communications
- Take Away
  - Smooth nonconvex nonlinear programs
  - Local superlinear/quadratic convergence rate

[18] **X. Dai** et al., "Distributed AC optimal power flow: A scalable solution for large-scale problems,", under review, 2026, [arXiv:2503.24086]

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Content

1. Introduction

2. Incremental Advances in Distributed Optimization for Nonlinear Programming

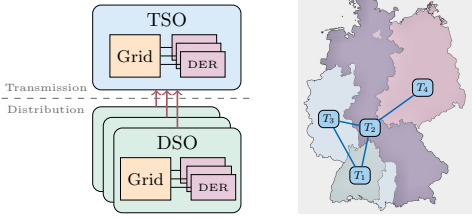
3. Barrier Augmented Lagrangian-based Alternating Direction Inexact Newton (BALADIN)

4. Flexibility Aggregation for Coordinating Integrated Transmission-Distribution (ITD) Systems

5. Conclusion & Future Work

Methodology

- Leverage the hierarchical structure



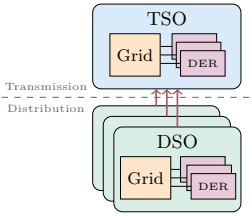
(a) Aggregating Flexibility (b) TSOs Coordination

Figure 15: Proposed real-time coordination of ITD systems [22]

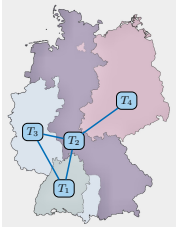
[22] **X. Dai** et al., "Real-time coordination of integrated transmission and distribution systems: Flexibility modeling and distributed NMPC scheduling," *Electric Power Systems Research*, Sepecial issue from Power System Computation Conference (PSCC), 2024

Methodology

- Leverage the hierarchical structure
  - Decouple transmission and distribution networks



(a) Aggregating Flexibility



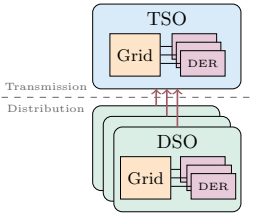
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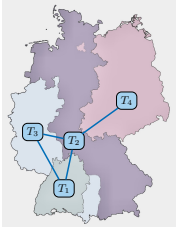
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  - DSOs provide temporal decoupled feasible operation sets



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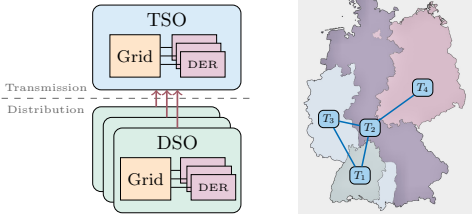
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Methodology

- Leverage the hierarchical structure
  - Decouple transmission and distribution networks
  - DSOs provide temporal decoupled feasible operation sets
  - TSOs determine multiperiod OPF via distributed optimization



(a) Aggregating Flexibility (b) TSOs Coordination

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Flexibility Aggregation within ITD Systems

- Hierarchical graph abstraction of ITD systems

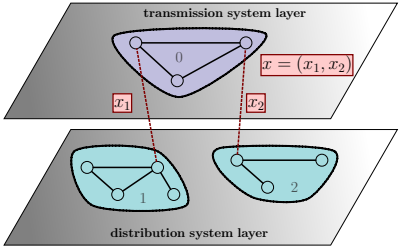


Figure 16: Hierarchical graph of an ITD system with $\mathcal{D}=\{1,2\}$

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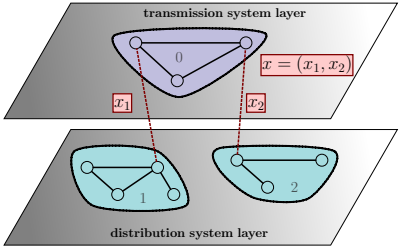


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(10a)

(10b)

(10c)

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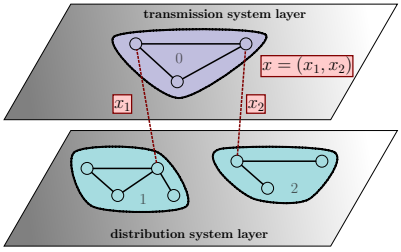


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$$\min_{x,y} \quad f_0(x, y_0) \tag{10a}$$

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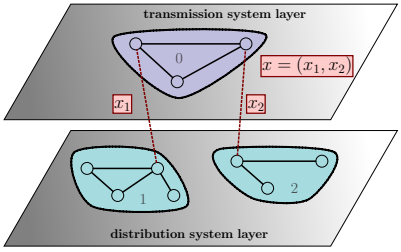


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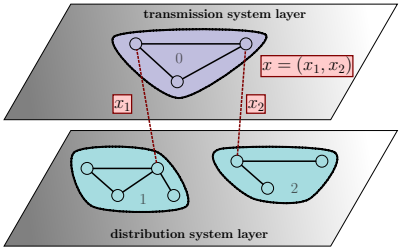


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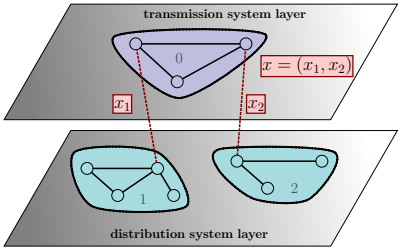


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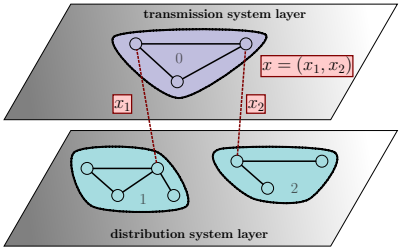


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systems ℓ:

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LinDistFlow for Flexibility Aggregation

Remark (Radial Network)

The DistFlow model serves as an exact relaxation of the nonconvex AC power flow equations in radial networks by eliminating voltage angle dependencies.

Remark (Lossless Approximation)

The LinDistFlow model is a linearized approximation of the DistFlow model, derived under the assumption of lossless power lines, i.e., neglecting line losses.

[25] M. Farivar et al., "Branch flow model: Relaxations and convexification (Parts I, II)," *IEEE Trans. Power Syst.*, 2013

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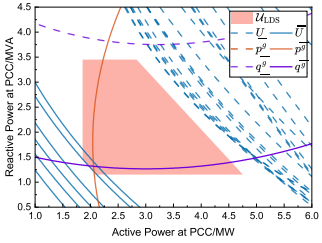


Figure 17: Flexibility mismatch by using LinDistFlow model. The solid and dashed curves give upper and lower limits on squared voltage magnitudes (blue), active power (red), and reactive power (purple) of DERs [24]

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Loss Compensation for LinDistFlow

Enhenced Feasibility Region

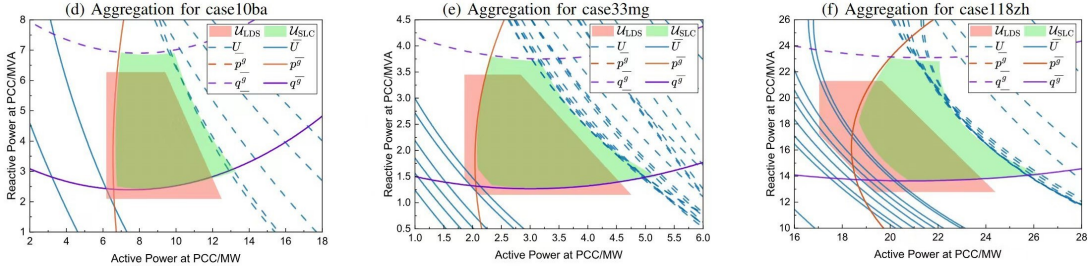


Figure 18: Enhanced flexibility aggregation using LinDistFlow model with loss compensation

[24] Y. Jiang, **X. Dai**[§] et al., "Enhanced flexibility aggregation using LinDistFlow model with loss compensation," in *IEEE PowerTech*, 2025, [§] **corresponding**

Loss Compensation for LinDistFlow

Enhenced Optimality

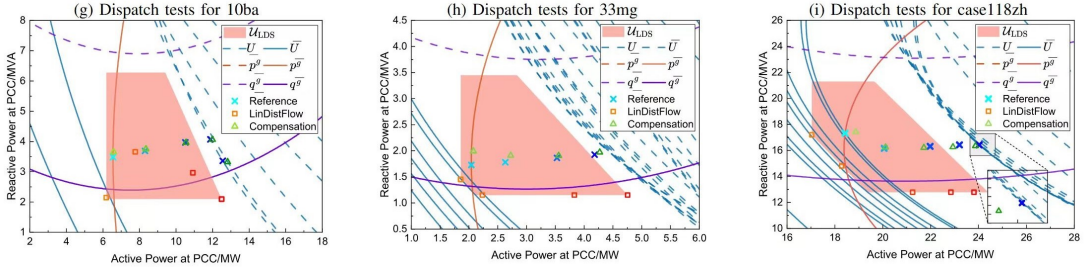


Figure 19: Optimal solutions to the aggregated AC OPF problems

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Nonlinear Flexibility Aggregation

Predictor-Corrector Methods for General Nonlinear Model (Meshed)

- Nonlinear systems

$$g(x, y) = 0,\ x \in \mathcal{X} \subseteq \mathbb{R}^m,\ y \in \mathcal{Y} \subseteq \mathbb{R}^n,$$

with $g:,\mathbb{R}^m \times \mathbb{R}^n \to \mathbb{R}^n$ and the implicit feasible set

$$\mathcal{X}^{\text{imp}} = \{x \in \mathcal{X} \mid \exists\, y \in \mathcal{Y} \text{ such that } g(x, y) = 0\}$$

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- Predictor-Corrector Approximation

$$\hat{y}(x) := y_0 - \underbrace{M_0^{-1} \partial_x g(x_0, y_0)(x - x_0)}_{\text{predictor}} - \underbrace{M_0^{-1}\, g(x, \bar{y}(x))}_{\text{corrector}}$$

with

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Nonlinear Flexibility Aggregation

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- Approximated implicit feasible set

$$\hat{\mathcal{X}}^{\text{imp}} = \{x \in \mathcal{X} \mid\ \hat{y}(x) \in \mathcal{Y}\}$$

Nonlinear Flexibility Aggregation

Predictor-Corrector Methods for General Nonlinear Model (Meshed)

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$$\hat{y}(x) := y_0 - \underbrace{M_0^{-1} \partial_x g(x_0, y_0)(x - x_0)}_{\text{predictor}} - \underbrace{M_0^{-1}\, g(x, \bar{y}(x))}_{\text{corrector}}$$

with

$$M_0 := \partial_y g(x_0, y_0), \text{ and } \bar{y}(x) := y_0 - M_0^{-1} \partial_x g(x_0, y_0)\,(x - x_0)$$

- Approximated implicit feasible set

$$\hat{\mathcal{X}}^{\text{imp}} = \{x \in \mathcal{X} \mid\ \hat{y}(x) \in \mathcal{Y}\}$$

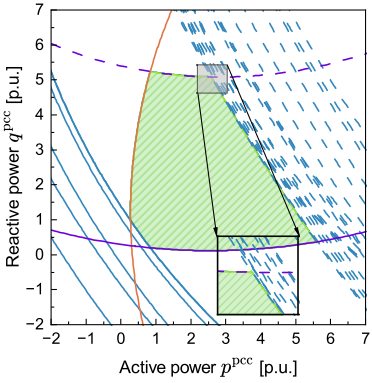


Figure 20: Illustrative comparison of the proposed approximated set $\hat{\mathcal{X}}^{\text{imp}}$ vs. the exact feasible regions $\mathcal{X}^{\text{imp}}$. The solid and dashed curves give upper and lower limits on squared voltage magnitudes (blue), active power (red), and reactive power (purple) of DERs.

Nonlinear Flexibility Aggregation

Approximation Error Analysis

- Implicit feasible set:
- Predictor-Corrector Approximation:
- Approximated implicit feasible set:

$$\mathcal{X}^{\text{imp}} := \{x \in \mathcal{X} \mid \exists\, y \in \mathcal{Y} \text{ such that } g(x,y) = 0\}$$

$$\hat{\mathcal{X}}^{\text{imp}} := \{x \in \mathcal{X} \mid \hat{y}(x) \in \mathcal{Y}\}$$

$$\hat{y}(x) := y_0 - M_0^{-1}\partial_x g(x_0, y_0)(x - x_0) - M_0^{-1}\, g(x, \bar{y}(x))$$

[23] **X. Dai** et al., "Distributed coordination for transmission-distribution systems with nonlinear flexibility aggregation,", under review, 2026, [arXiv:2608.06341]

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Theorem (State Mismatch [23, Thm. 2])

Suppose g is continuously differentiable, and the based-point Jacobian $M_0 = \partial_y g(x_0, y_0)$ is nonsingular. Then there exist neighborhoods $\mathcal{U} \subseteq \mathcal{X}$ such that the predictor-corrector approximation $\hat{y}(x)$ satisfies

$$\|\hat{y}(x) - y^\star(x)\| \;\le\; C\, \|x - x_0\|^3, \qquad \forall x \in \mathcal{U},$$

with $y^\star(x)$ denoting the exact local state with respect to the coupling x.

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with $y^\star(x)$ denoting the exact local state with respect to the coupling x.

Corollary (Flexibility Set Mismatch [23, Col. 1])

Suppose the assumptions of Theorem 10 hold. Then, for all $x \in \mathcal{U}$, the following conditons are satisfied:

$$\operatorname{dist}(y^\star(x),\, \mathcal{Y}) \le C\, \|x - x_0\|^3,\ \forall x \in \hat{\mathcal{X}}^{\text{imp}} \setminus \mathcal{X}^{\text{imp}}, \qquad \text{(False Flexibility)}$$

$$\operatorname{dist}(y^\star(x),\, \mathbb{R}^n \setminus \mathcal{Y}) \le C\, \|x - x_0\|^3,\ \forall x \in \mathcal{X}^{\text{imp}} \setminus \hat{\mathcal{X}}^{\text{imp}}, \qquad \text{(Lost Flexibility)}$$

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Nonlinear Flexibility Aggregation

Benchmarks on Flexibiltiy Set Mismatch

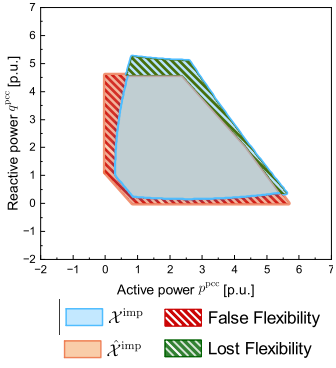


Figure 21: The benchmark metrics

Nonlinear Flexibility Aggregation

Benchmarks on Flexibiltiy Set Mismatch

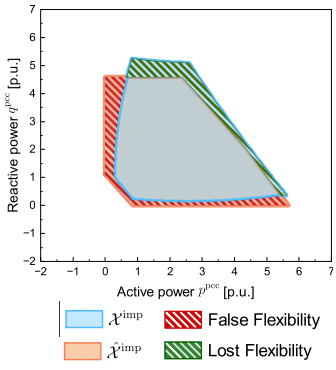


Figure 21: The benchmark metrics

Table 8: Approximation errors: false- & lost flexibility

Cases	Topology	False Flexibility [%]				Lost Flexibility [%]			
		LDS	CR	TP	PC	LDS	CR	TP	PC
case10ba	radial	5.14	161.02	4.22	0.04	5.63	0.00	2.12	0.04
...	...	...	...	...	...	...	...	...	...
case33bw	radial	5.22	171.12	4.77	0.03	6.99	0.00	2.42	0.00
case33bw	mesh	–	–	3.96	0.00	–	–	3.43	0.01
...	...	...	...	...	...	...	...	...	...
case118zh	radial	17.22	228.94	15.46	0.82	14.51	0.00	6.55	0.05
case118zh	mesh	–	–	7.07	0.05	–	–	3.90	0.03
...	...	...	...	...	...	...	...	...	...
case533mt_hi	radial	4.47	419.33	2.11	0.18	5.69	0.00	1.64	0.02
case533mt_hi	mesh	–	–	2.31	0.20	–	–	1.78	0.00
case533mt_lo	radial	5.26	465.99	2.08	0.37	2.81	0.00	1.64	0.00
case533mt_lo	mesh	–	–	2.15	0.25	–	–	1.94	0.01
KIT-CN	radial	3.61	48.34	0.93	0.04	3.86	0.00	1.01	0.01
KIT-CN	mesh	–	–	1.11	0.48	–	–	0.83	0.00

* LDS CR, TP, and PC denote LinDistFlow, convex DistFlow relaxation, tangential predictor, and the proposed predictor-corrector method, respectively.

Nonlinear Flexibility Aggregation

Coupling Choices and Temporal Decomposition

(a) Feasible set at PCC

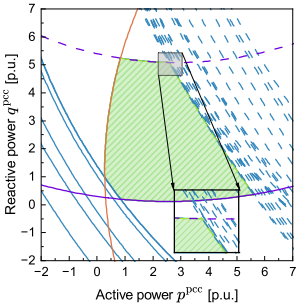


Figure 22: Implicit flexibiltiy set in different coupling domain.

Nonlinear Flexibility Aggregation

Coupling Choices and Temporal Decomposition

(a) Feasible set at PCC

(b) Feasible set at DER

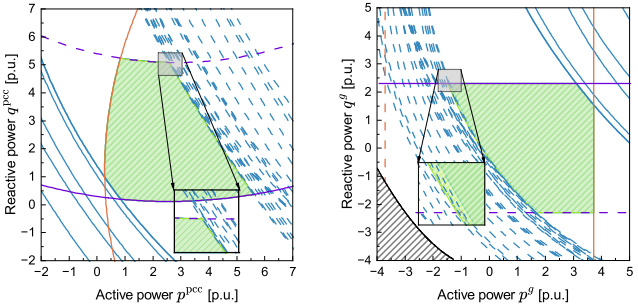


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Nonlinear Flexibility Aggregation

Coupling Choices and Temporal Decomposition

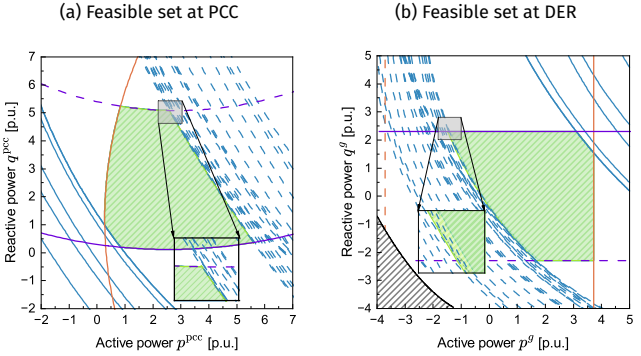


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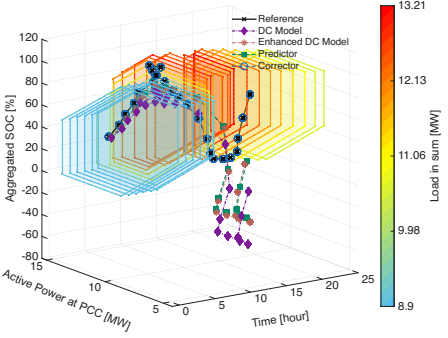


Figure 23: Temporal Deocmposition.

Nonlinear Flexibility Aggregation for T-D Systems

Multiperiod ITD with Spatio- and Temporal-Decomposition

- Settings
  - Multiple DSOs connected to one TSO
  - Meshed distribution systems
  - Multiperiod AC OPF
  - IPOPT as centralized nonlinear solver

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Table 9: Performance of centralized and aggregated approaches

TSO case	Approach	Aggregation [s]	Initialization [s]	Solve [s]	Total [s]
case14	Centralized [s]	–	2.12	4.94	7.06
	Aggregated [s]	0.15	0.27	0.73	1.15
	Speedup	–	7.85×	6.77×	6.14×
case118	Centralized [s]	–	37.33	20.01	57.34
	Aggregated [s]	0.15	4.11	4.16	8.42
	Speedup	–	9.08×	4.81×	6.81×

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Table 10: Solve Time: centralized vs. aggregated

Case	Approach	Centr.	Agg.	Total
case14	Iterations	31	19	1.63×
	Linear Solver [s]	3.65	0.56	6.52×
	Function Eval. [s]	0.75	0.02	37.5×
	Total [s]	4.94	0.73	6.77×
case118	Iterations	31	22	1.41×
	Linear Solver [s]	14.96	3.14	4.76×
	Function Eval. [s]	3.07	0.16	19.2×
	Total [s]	20.01	4.16	4.81×

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Nonlinear Flexibility Aggregation

The Source of theComputational Speedup

Table 11: Solve Time: centralized vs. aggregated

Case	Approach	Centr.	Agg.	Total
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	Total [s]	20.01	4.16	4.81×

Table 12: Solve Time: centralized vs. aggregated

Case	Structural metric	Centralized	Aggregated	Reduction
case14	No. of variables	1.67E4	1.68E3	89.91%
	No. of constraints	1.65E4	9.08E3	44.88%
	KKT matrix order	3.31E4	1.85E4	44.05%
	NNZ of KKT	2.83E5	5.48E4	80.62%
	KKT density	2.58E−4	1.60E−4	38.09%
case118	No. of variables	1.77E5	1.09E4	93.82%
	No. of constraints	1.75E5	9.18E4	47.39%
	KKT order	3.52E5	1.87E5	46.86%
	NNZ of KKT	3.20E6	5.56E5	82.62%
	KKT density	2.59E−5	1.59E−5	38.46%

[23] **X. Dai** et al., "Distributed coordination for transmission-distribution systems with nonlinear flexibility aggregation,", under review, 2026, [arXiv:2608.06341]

Content

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2. Incremental Advances in Distributed Optimization for Nonlinear Programming

3. Barrier Augmented Lagrangian-based Alternating Direction Inexact Newton (BALADIN)

4. Flexibility Aggregation for Coordinating Integrated Transmission-Distribution (ITD) Systems

5. Conclusion & Future Work

Conclusion & Outlook

- Distributed Frameworks in Power Systems
- Ongoing Work

Conclusion & Outlook

- Distributed Frameworks in Power Systems

Table 13: Comparison: Distributed vs. Aggregated

	Barrier ALADIN	Flexibility Aggregation
Topology	General	Hierarchical
Coordination	Distributed	Leader-Follower
Methodolgy	Condense local KKT	Presolve locally
Communication	Iterative	Non-iterative
Accuracy	High	Moderate
Coupling	Dual	Primal
Scalability	5×10^5	2×10^5
Speedup	$2-4$ times	$5-8$ times

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  - GPU-enabled acceleration for large-scale energy systems
  - Fast maximum loadability screening, e.g., for data center / DERs integration studies
  - Numerical stability for large-scale capacity expansion model (MILP).
  - Sensitivity analysis of the distribution systems under reverse power flow conditions.

References

[1] A. Murray et al., “On grid partitioning in acli2024bess optimal power flow,” in *2020 IEEE PES ISGT-Europe*, IEEE, 2020

[2] W. Zheng et al., “A fully distributed reactive power optimization and control method for active distribution networks,” *IEEE Trans. Smart Grid*, 2015

[3] Q. Peng et al., “Distributed optimal power flow algorithm for radial networks,” *IEEE Trans. Smart Grid*, 2016

[4] A. Rajaei et al., “Decentralized transactive energy management of multi-microgrid distribution systems based on admm,” *Int. J. Electr. Power Energy Syst.*, 2021

[5] T. Erseghe, “Distributed optimal power flow using ADMM,” *IEEE Trans. Power Syst.*, 2014

[6] J. Guo et al., “A case for nonconvex distributed optimization in large-scale power systems,” *IEEE Trans. Power Syst.*, 2017

[7] S. Mhanna et al., “Adaptive ADMM for distributed AC optimal power flow,” *IEEE Trans. Power Syst.*, 2018

[8] K. Sun et al., “A two-level distributed algorithm for nonconvex constrained optimization,” *Computational Optimization and Applications*, 2023

[9] A. Engelmann et al., “Toward distributed OPF using ALADIN,” *IEEE Trans. Power Syst.*, 2018

[10] N. Meyer-Huebner et al., “Distributed optimal power flow in hybrid ac–dc grids,” *IEEE Trans. Power Syst.*, 2019

[11] A. Engelmann et al., “Decomposition of nonconvex optimization via bi-level distributed ALADIN,” *IEEE Trans. Control Netw. Syst.*, 2020

[12] B. Houska et al., “An augmented lagrangian based algorithm for distributed nonconvex optimization,” *SIAM Journal on Optimization*, 2016

[13] J. Zhai§, **X. Dai**§ et al., “Distributed optimal power flow for VSC-MTDC meshed AC/DC grids using ALADIN,” *IEEE Trans. Power Syst.*, 2022, §**equally**

[14] **X. Dai** et al., “Rapid scalable distributed power flow with open-source implementation,” in *9th IFAC Conference on Networked Systems*, 2022

[15] Y. Jiang§, **X. Dai**§ et al., “Global ALADIN for distributed nonconvex optimization,”, 2026, under review, §**equally**

[16] **X. Dai** et al., “Advancing distributed AC optimal power flow for integrated transmission-distribution systems,” *IEEE Trans. Netw. Sci. Eng.*, 2025

[17] **X. Dai** et al., “Ensuring data privacy in AC optimal power flow with a distributed co-simulation framework,” *Electric Power Systems Research*, Sepecial issue from Power System Computation Conference (PSCC), 2024

[18] **X. Dai** et al., “Distributed AC optimal power flow: A scalable solution for large-scale problems,”, under review, 2026, [arXiv:2503.24086]

[19] J. Nocedal et al., *Numerical optimization*. Springer Science & Business Media, 2006

References

[20] N. I. Gould, “On practical conditions for the existence and uniqueness of solutions to the general equality quadratic programming problem,” *Mathematical Programming*, 1985

[21] E. V. Haynsworth, “Determination of the inertia of a partitioned hermitian matrix,” *Linear algebra and its applications*, 1968

[22] **X. Dai** et al., “Real-time coordination of integrated transmission and distribution systems: Flexibility modeling and distributed NMPC scheduling,” *Electric Power Systems Research*, Sepecial issue from Power System Computation Conference (PSCC), 2024

[23] **X. Dai** et al., “Distributed coordination for transmission-distribution systems with nonlinear flexibility aggregation,”, under review, 2026, [arXiv:2608.06341]

[25] M. Farivar et al., “Branch flow model: Relaxations and convexification (Parts I, II),” *IEEE Trans. Power Syst.*, 2013

[26] M. E. Baran et al., “Network reconfiguration in distribution systems for loss reduction and load balancing,” *IEEE Transactions on Power Delivery*, 1989

[24] Y. Jiang, **X. Dai**[§] et al., “Enhanced flexibility aggregation using LinDistFlow model with loss compensation,” in *IEEE PowerTech*, 2025,[§] **corresponding**

Acknowledgments

- Academic Partners



- Industry Collaborators

